Wastewater analysis using R

Introduction

In this module, we will analyze a wastewater dataset from the CDC to investigate how COVID-19 viral levels change through time.

We will learn ways to manipulate dates in R using the library **lubridate** (part of the Tidyverse package).

Recap: Dates and times in R

There are two most popular R classes used when working with dates and times:

- `Date` class representing a calendar date and
- `POSIXct` class representing a calendar date with hours, minutes, seconds.
- A date-time object in R is a point on the timeline stored as the number of seconds since *January 1st, 1970, 00:00 UTC*.
- This date is also called the **Unix epoch** and serves as a reference by computer systems to track and operate with time.

`lubridate` package:

- The lubridate package allows us to work with dates and times in a simple and efficient way.
- In this module, we will learn to:
  - Input dates in different formats,
  - Perform basic operations with dates, and
  - Extract components of dates.

`lubridate` is very flexible for inputting dates in different formats:

- `YYYY\-MM\-DD`:

```
a <- ymd("2024-09-20")
```

- `DD-MM-YYYY`:

```
a <- dmy("20-09-2024")
```

- `MM-DD-YY`:

```
a <- dmy("20-09-2024")
```

Extract the components of a date with `lubridate`:

- Year, month, day of the month (`mday`):

```
a <- ymd_hms("2024-09-20 3:59:01")
year(a)
```

```
[1] 2024
```

```
month(a)
```

```
[1] 9
```

```
mday(a)
```

```
[1] 20
```

Extract the components of a date continued:

- Day of the year (`yday`), day of the week (`wday`), hour, minute and seconds

```
yday(a)

[1] 264

wday(a)

[1] 6

hour(a)

[1] 3

minute(a)

[1] 59

second(a)

[1] 1
```

Perform operations with dates:

```
a <- ymd("2024-09-20")
b <- ymd("2025-07-15")
b - a
```

```
Time difference of 298 days
```

```
a - ymd("2024-08-20")
```

```
Time difference of 31 days
```

```
year(a) - 1989
```

```
[1] 35
```

GUT CHECK:

What does the package `lubridate` allows us to do?

- Work with dates in R
- Produce time series data
- Plot time series data

Data used:

We will use data reporting the level of SARS-CoV-2 (the virus responsible of COVID-19) viral particles in wastewater.

This data reports the level of SARS-CoV-2 at the state, regional and national level through time. According to this website, National, regional, and state/territory data represent the median values across all wastewater treatment plants in the respective area.

You can read about the data and how it was collected here: https://www.cdc.gov/nwss/rv/COVID19-statetrend.html

Download the data

We will use the data located here:

```
dat <- read_csv("http://daseh.org/data/wastewater_COVID-19_State_and_Territory_Trends.csv")
```

Clean up the data a little bit:

- Some column names contain `/` and spaces, which are not standard form. We will change them so it is easier to work with them.To do so, we use the function `clean_names` part of `janitor` package. Look at the column names of the old dataset (called `dat`) and the column names of the new dataset (called `dat1`), the new names are all in lower case with underscores between words.

```
dat1 <- dat |> janitor::clean_names()
colnames(dat)
```

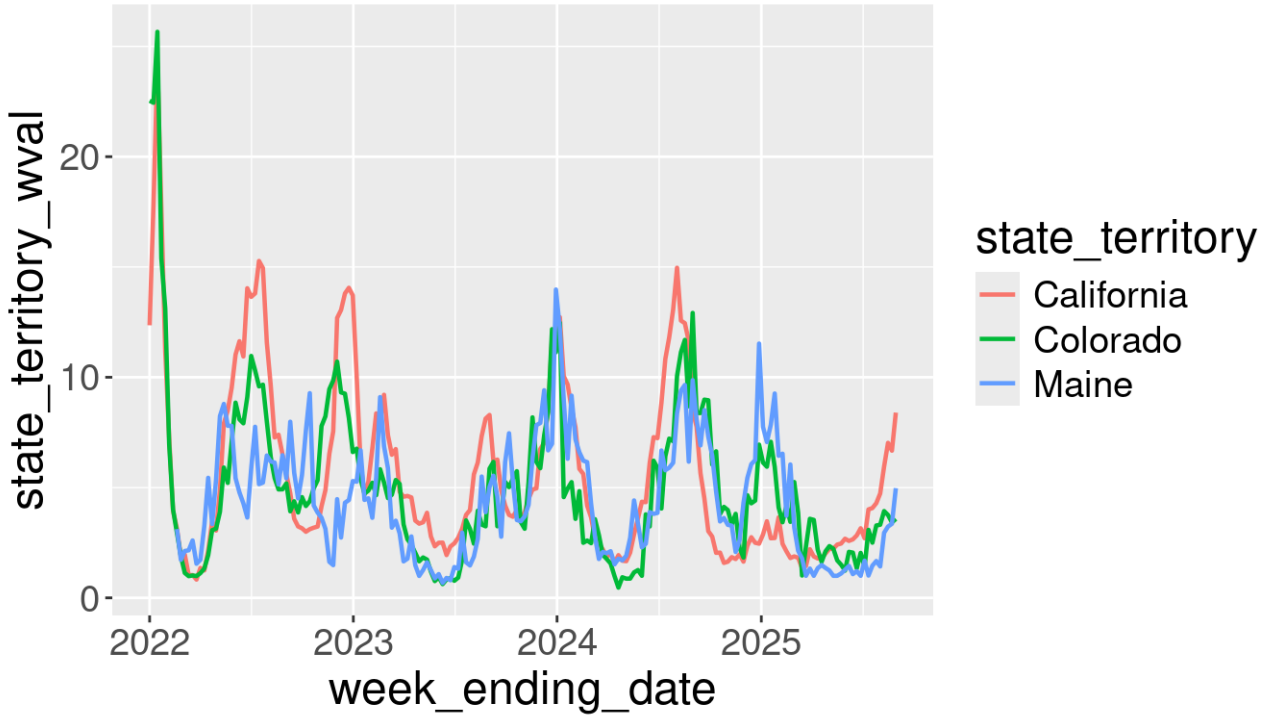
```
[1] "State/Territory"        "Week_Ending_Date"       "Data_Collection_Period"
[4] "State/Territory_WVAL"   "National_WVAL"          "Regional_WVAL"
[7] "WVAL_Category"          "Coverage"               "date_updated"
```

```
colnames(dat1)
```

```
[1] "state_territory"        "week_ending_date"       "data_collection_period"
[4] "state_territory_wval"   "national_wval"          "regional_wval"
[7] "wval_category"          "coverage"               "date_updated"
```

Mission: Understand the trends of COVID-19 disease through time in different parts of the United States (US).

Let's imagine we want to understand how SARS-CoV-2 infections (and hence COVID-19 disease) change through time in three different parts of the US: the East coast, the West coast, and the Midwest. We want to obtain plots that should look like this:



Clean up the data (continued)

- We will not be using the columns `Coverage` and `date_updated` so we will remove them from our data frame.
- We will conduct our analysis using all the results available, so that we will remove the rows with results at 1 year or every 6 months.

```
dat2 <- dat1 |> select(!c(coverage, date_updated)) |> filter(data_collection_period == "All Results")
head(dat2)
```

```
# A tibble: 6 × 7
  state_territory week_ending_date data_collection_period state_territory_wval
  <chr>           <date>           <chr>                                  <dbl>
1 Tennessee       2022-04-02       All Results                            0.987
2 Tennessee       2022-02-12       All Results                            2.45
3 Tennessee       2022-01-01       All Results                            3.37
4 Tennessee       2022-01-08       All Results                            1.31
5 Wyoming         2023-09-23       All Results                            3.94
6 Vermont         2022-05-07       All Results                           13.9
# ℹ 3 more variables: national_wval <dbl>, regional_wval <dbl>,
#   wval_category <chr>
```

Select 3 states to look at:

- We will select three states in the US in three different parts (East, Center, West) to look at their wastewater data: South Carolina, Nebraska, Washington.

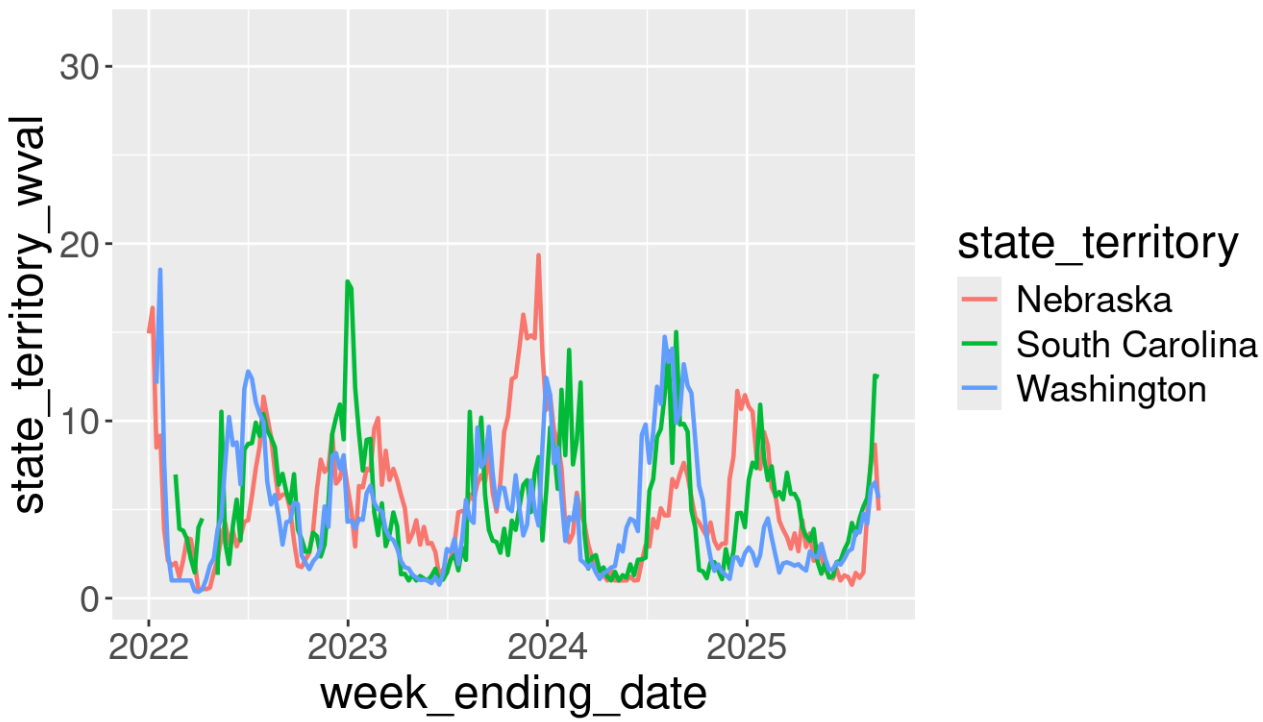
```
df <- dat2 |> filter(state_territory %in% c('South Carolina', 'Nebraska', 'Washington'))
head(df)
```

```
# A tibble: 6 × 7
  state_territory week_ending_date data_collection_period state_territory_wval
  <chr>           <date>           <chr>                                  <dbl>
1 South Carolina  2024-02-24       All Results                             8.89
2 South Carolina  2024-03-16       All Results                             1.67
3 South Carolina  2025-01-25       All Results                            10.9
4 South Carolina  2023-12-30       All Results                             5.99
5 South Carolina  2024-04-06       All Results                             1.53
6 South Carolina  2024-06-01       All Results                             1.91
# ℹ 3 more variables: national_wval <dbl>, regional_wval <dbl>,
#   wval_category <chr>
```

Look at the data!

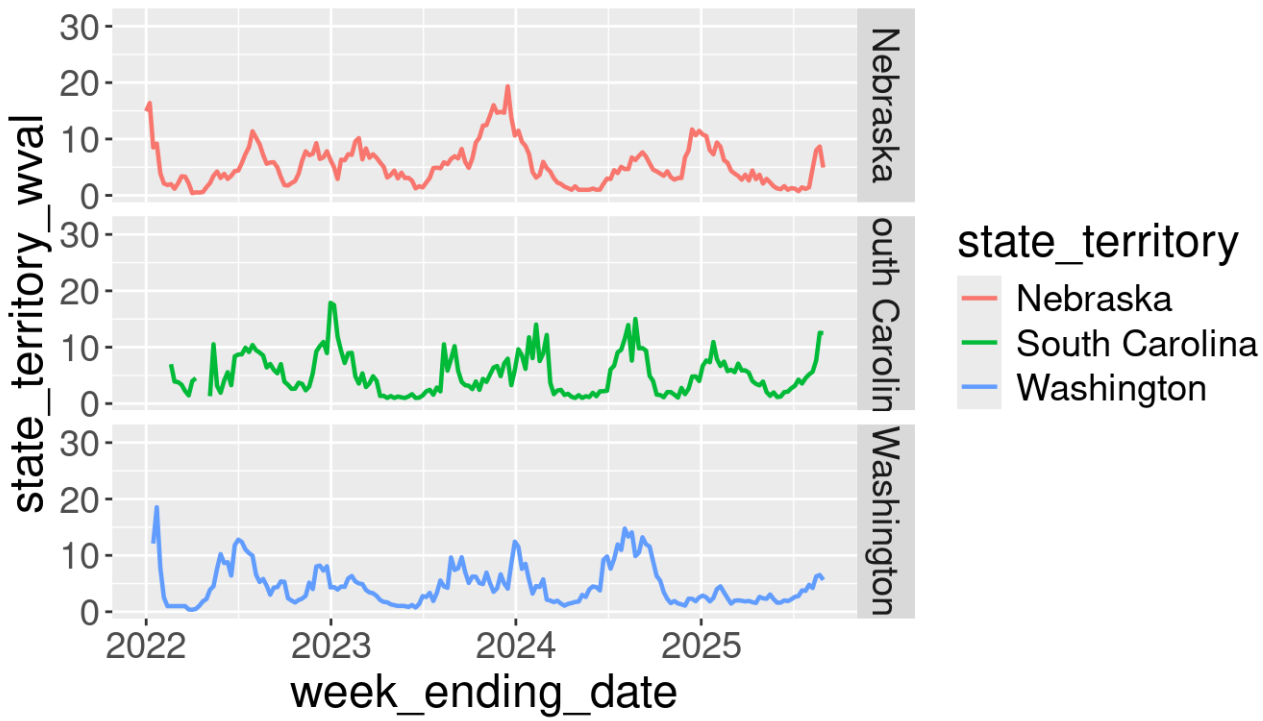
We will plot the data using ggplot:

```
ggplot(data=df, aes(x = week_ending_date, y=state_territory_wval,
                 color=state_territory)) + geom_line(linewidth=0.8) +
  theme(text = element_text(size = 18))
```



Split the plots to see the patterns more clearly:

```
ggplot(data=df, aes(x = week_ending_date, y=state_territory_wval, color=state_territory)) +
  geom_line(linewidth=0.8) + theme(text = element_text(size = 18)) + facet_grid(state_territory~.)
```



What can we say about this data?

- Seasonality pattern for each state.
- Looks like the peaks of the 2023-2024 winter season occurred at slightly different times in each state.
- Most of the big waves happen in winter time, but all three states had a summer outbreak in 2024.

Zoom in the data!

- Now, we want to investigate how big was the summer outbreak in each state in 2024.
- We use `lubridate` to select only the data from year 2024:

Quick quiz

Do you remember how to extract the year from a date called `mydate`?

1. `year(mydate)`
2. `extract(year, mydate)`
3. `mydate.year()`

Extract the data for year 2024:

```
df2024 <- df |> filter(year(week_ending_date) == 2024)
head(df2024)
```

```
# A tibble: 6 × 7
  state_territory week_ending_date data_collection_period state_territory_wval
  <chr>           <date>           <chr>                                <dbl>
1 South Carolina  2024-02-24       All Results                           8.89
2 South Carolina  2024-03-16       All Results                           1.67
3 South Carolina  2024-04-06       All Results                           1.53
4 South Carolina  2024-06-01       All Results                           1.91
5 South Carolina  2024-12-28       All Results                           4.01
6 South Carolina  2024-04-27       All Results                           1
# ℹ 3 more variables: national_wval <dbl>, regional_wval <dbl>,
#   wval_category <chr>
```

Obtain the peak viral level in each state.

First, we need to group the data by state, we use the function`group_by` from `dplyr` to do it:

```
maxRows <- df2024  |> group_by(state_territory) |>
```

then retrieve the rows with the peak level:

```
maxRows <- df2024  |> group_by(state_territory) |> ???
```

How do we do this?

Quick quiz

Do you remember what function to use to obtain the peak level?

1. `max(dat)`
2. `min(dat)`
3. `unique(dat)`

Obtain the peak viral level in each state.

To do this, we need to group the data by state and retrieve the rows with the maximum level:

```
maxRows <- df2024  |> group_by(state_territory) |>
          filter(state_territory_wval == max(state_territory_wval, na.rm=TRUE))
maxRows
```

```
# A tibble: 3 × 7
# Groups:   state_territory [3]
  state_territory week_ending_date data_collection_period state_territory_wval
  <chr>           <date>           <chr>                                  <dbl>
1 Washington      2024-08-03       All Results                             14.7
2 South Carolina  2024-08-24       All Results                             15.0
3 Nebraska        2024-12-14       All Results                             11.7
# ℹ 3 more variables: national_wval <dbl>, regional_wval <dbl>,
#   wval_category <chr>
```

Extract the peak wastewater levels for each state:

We will compute the date of the summer peak in each state:

```
Nebraska_peak <- maxRows %>% filter(state_territory=="Nebraska") %>%
                             pull(week_ending_date)
South_Carolina_peak <- maxRows %>% filter(state_territory=="South Carolina") %>%
                                   pull(week_ending_date)
Washington_peak <- maxRows %>% filter(state_territory=="Washington") %>%
                               pull(week_ending_date)
Nebraska_peak
```

```
[1] "2024-12-14"
```

```
South_Carolina_peak
```

```
[1] "2024-08-24"
```

```
Washington_peak
```

```
[1] "2024-08-03"
```

Compute the number of days between the peak wastewater levels for each state during the summer wave:

- Difference between South Carolina and Nebraska

```
Nebraska_peak - South_Carolina_peak
```

```
Time difference of 112 days
```

- Difference between South Carolina and Washington

```
South_Carolina_peak - Washington_peak
```

```
Time difference of 21 days
```

- Difference between Nebraska and Washington

```
Nebraska_peak - Washington_peak
```

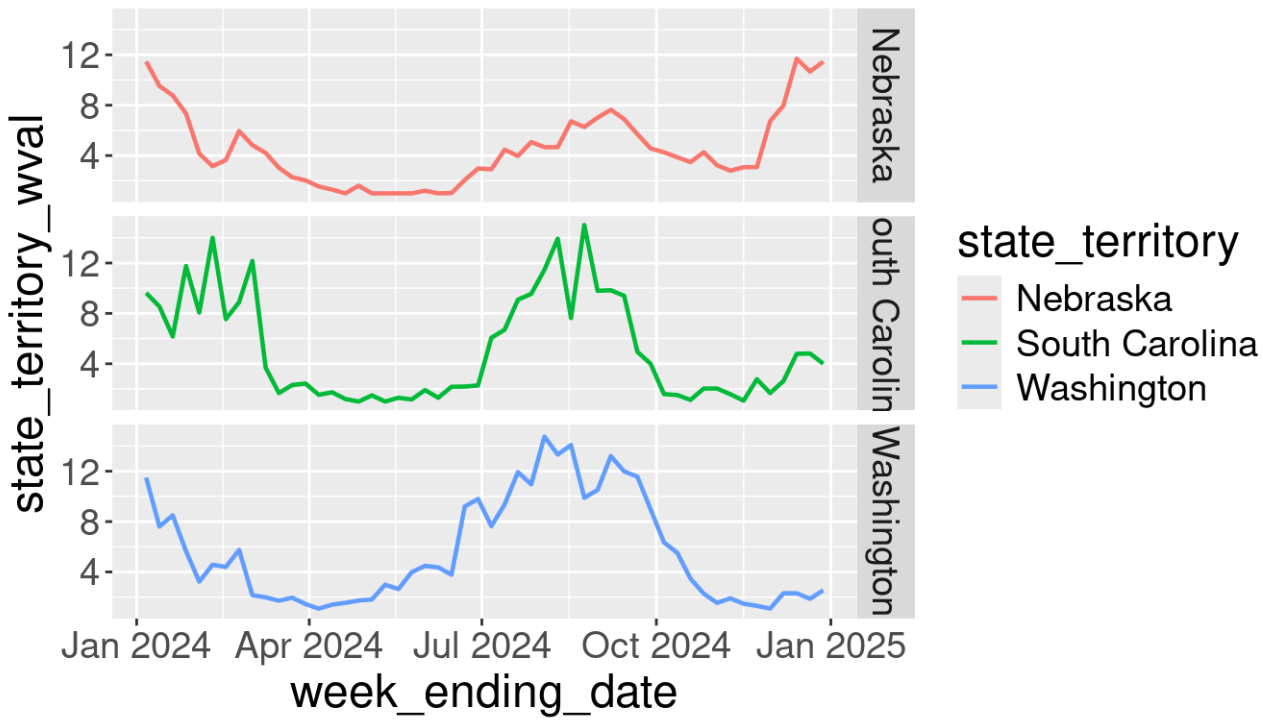
```
Time difference of 133 days
```

Summer wave:

- From this data, we can see that South Carolina and Washington had a summer wave that peaked 21 days apart.
- However, the maximum calculated for Nebraska corresponds to the winter wave, not to the summer wave.

Let's look again at the plot, but this time restricted to the year 2024:

```
ggplot(data=df2024, aes(x = week_ending_date, y=state_territory_wval, color=state_territory)) +
  geom_line(linewidth=0.8) + theme(text = element_text(size = 18)) + facet_grid(state_territory~.)
```



Summer wave continued:

How can we obtain the day that the wastewater measurement reached the maximum level for Nebraska in the summer?

Summer wave continued:

We will further zoom our attention to the summer months and modify the code to look at the data for the months of June, July and August 2024:

```
maxRows <- df2024 |> filter(month(week_ending_date) %in% c(06, 07, 08)) |>
                      group_by(state_territory) |>
                      filter(state_territory_wval == max(state_territory_wval, na.rm=TRUE))

maxRows
```

```
# A tibble: 3 × 7
# Groups:   state_territory [3]
  state_territory week_ending_date data_collection_period state_territory_wval
  <chr>           <date>           <chr>                                  <dbl>
1 Washington      2024-08-03       All Results                            14.7
2 South Carolina  2024-08-24       All Results                            15.0
3 Nebraska        2024-08-31       All Results                             7.01
# ℹ 3 more variables: national_wval <dbl>, regional_wval <dbl>,
#   wval_category <chr>
```

Number of days between summer peaks

Now we can compute the number of days between the peak waste water levels for each state during the summer wave:

```
Nebraska_peak <- maxRows %>% filter(state_territory=="Nebraska") %>% pull(week_e
South_Carolina_peak <- maxRows %>% filter(state_territory=="South Carolina") %>%
Washington_peak <- maxRows %>% filter(state_territory=="Washington") %>% pull(we
```

- Difference between South Carolina and Nebraska

```
Nebraska_peak - South_Carolina_peak
```

```
Time difference of 7 days
```

- Difference between South Carolina and Washington

```
South_Carolina_peak - Washington_peak
```

```
Time difference of 21 days
```

Conclusion

- South Carolina and Washington had a summer wave that peaked 21 days apart.
- Nebraska and Washington had a summer wave that peaked 28 days apart.
- Nebraska and South Carolina had a summer wave that peaked one week apart.